

RESEARCH ARTICLE

Avoiding misleading estimates of among-individual variance caused by non-random sampling of individuals in a changeable environment

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Abstract

1. Animal ecologists frequently quantify variance in hierarchically structured traits in wild populations. Importantly, phenotypic plasticity within the period of measurement can modify the trait of interest in response to various unmeasured, temporally or spatially changeable, environmental conditions. Non-random sampling among units of the random effect (e.g. individuals) regarding the environment at issue may lead to estimates of the variance among ($\hat{\sigma}_I^2$) or within ($\hat{\sigma}_W^2$) such units that conflate several types of processes. This mixing of underlying biology can affect interpretations of the random effect variance. Here, we explore the conditions leading to this situation and assess potential solutions when relevant information is missing.
2. We simulated a trait's phenotypic values that depended on the environmental variable, and individuals that differed in their deviation to the mean population phenotype (random intercepts). We also simulated different types of variation in an environmental variable that was either shared or specific to each individual. We then varied the repeatability in the timing of sampling (R_{IS}^2) and analysed simulated datasets using linear mixed-effect models with different fixed- and random-effect structures.
3. In the presence of unmeasured environmental factors, the estimated among-individual variance ($\hat{\sigma}_I^2$) contained a larger signature of the current environment as

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the strength of the temporal autocorrelation and the repeatability in the timing of sampling (R_{IS}^2) increased. For low to moderate values of R_{IS}^2 (e.g. <60% of the total variance in our simulations) the risk of pre-study and within-study effects conflating estimates of variance components was low and could easily be corrected with a model including period or individual-period combination as random effects. Higher R_{IS}^2 led to an increase in conflating effects that were difficult to correct.

4. Our study shows the importance of limiting the variance among individuals in the timing structure of sampling (R_{IS}^2). We recommend researchers estimate R_{IS}^2 and report it in papers. Finally, R_{IS}^2 can be limited by sampling all individuals in the same period, or sensitivity analyses could be conducted by removing extreme sampling dates at the analysis stage to reduce R_{IS}^2 .

KEYWORDS

biased sampling, conflation of effects, heritability, linear mixed-effect models, repeatability, temporal autocorrelation, temporal trends, unmeasured environmental effects

1 | INTRODUCTION

Mixed-effect models have become increasingly popular for studying a variety of ecological and evolutionary questions. They are used either to control for pseudoreplication issues related to non-independent measurements when estimating fixed effects, or as a tool for decomposing variance components and analysing the hierarchical structure of complex traits (Downs & Dochtermann, 2014; Gimenez & Choquet, 2010). These models elegantly estimate among-subject variance and repeatability, an index of the strength of individual relative consistency in a trait over time and individual phenotypic plasticity (Dingemanse et al., 2010). Because of their robustness to unbalanced sampling designs and their ability to analyse hierarchically structured data, mixed-effect models have become the primary statistical tool used for studies of individual differences (e.g. animal personality) (Dingemanse et al., 2010; Réale et al., 2007), multilevel associations between traits (Dingemanse & Dochtermann, 2013), phenotypic plasticity (Husby et al., 2010; Martin & Réale, 2008; Nussey et al., 2005, 2007), senescence and the costs of reproduction (Berger et al., 2015; Martin & Festa-Bianchet, 2011; Wilson et al., 2008), phenological responses to climate change (Brommer et al., 2005), quantitative genetics (Kruuk, 2004; Kruuk & Hadfield, 2007; Wilson et al., 2010) and selection (Dingemanse et al., 2021; Hayward et al., 2014; Santostefano et al., 2019). While several studies have assessed how sampling designs can affect the power of these analyses (Martin et al., 2011; van de Pol, 2012), less is known about the effect of sampling design choices on key estimators. Here we study the impact of repeatability in the timing of sampling on among-group variance and its interpretation, particularly when groups (e.g. individuals) plastically respond to unknown environmental factors during the sampling period. We also assess several methods by which researchers might address this issue.

Mixed effects models have three general terms of potential relevance. As with all linear models, predictors (fixed effects) explain variance in observations collected during the study. If such observations are repeated within certain units (i.e. individuals), then a mixed model has a random effect (individual identity) and a residual variance (e.g. within-individual variance). Among-individual variance corresponds to the variance of individual deviations from the trait mean, while residual variance represents the variation in trait measurements taken from individuals over the period of data collection that cannot be explained by fixed or random effects of the model. Together, these terms account for variance unexplained by fixed effects and for processes other than those specified in the model.

Variance components estimated in mixed models can be used to test biological hypotheses. The biology studied is therefore decisive for the interpretation of these terms and the alternative processes likely to affect these estimates. For example, researchers assess the variance due to genetic or early developmental conditions that have long-lasting influences on organismal traits. Any process occurring before the measurement period could influence trait variance during the measurement period, either by affecting the mean phenotype or by interacting with the conditions during the measurement period to produce plasticity variance. In this sense, many studies aim to use mixed models to retrieve the variance due to processes that influenced subjects *before* coming into the study, as opposed to variance occurring *during* the study period. The random effect is then often interpreted as measuring pre-study effects. A major problem is that sampling designs may translate processes that occur within the study into the variance components, thereby aggregating within-study processes with before-study ones. It is important to understand when sampling bias occurs, as it affects conclusions about sources of variance among individuals or, more generally, variances at other hierarchical levels. Moreover, sampling designs and mixed-effect models that disregard important random effects may

cause part of a given variance component (e.g. residual variance) to be transferred to the among-individual variance components (Schielzeth et al., 2020). Therefore, variance in sampling may also affect estimates of proportions such as repeatability or heritability.

Sometimes sampling different individuals at different time-points within this study period is unavoidable. For example, studies of breeding behaviour often sample different individuals at various stages of the breeding season, due to variations in phenology. Also, the phenotype of the individuals for the trait studied can affect the probability of captures throughout the study period and therefore the temporal distribution of measurements for each individual (Biro & Dingemanse, 2009). For example, bold or more active individuals may be trapped earlier in the season than shy or less active ones. Males and females or young individuals may show different activity cycles during the season. Moreover, all studies fail to measure all the important processes affecting the focal phenotype. Typically, we assume the missing part goes into the residual variance and is uncorrelated with any other aspect of the model, but important unmeasured environmental effects may vary temporally or spatially over the study period. The combination of temporal environmental effects and temporal variability in sampling could lead to an estimation of among-individual variance that incorporates both pre-study and within-study effects (Figure 1). Our conclusions about the biology of such variation may then be mistaken (Niemelä & Dingemanse, 2017; Réale & Dingemanse, 2012; Wilson, 2018).

Environments can vary temporally or spatially with different patterns (Figure 2). Phenological or seasonal changes or altitudinal or latitudinal variation in the environment can be linear and directional. Circadian or circannual changes are more cyclical. These directional trends or cyclical patterns come with temporal autocorrelation. Autocorrelation in environmental variables is ubiquitous and affects many ecological patterns (Dormann et al., 2007; Hamel et al., 2012; Mitchell et al., 2019). Other environmental changes may be unpredictable and thus may be stochastic from the perspective of the subject. For example, the risk of predation may fluctuate for external reasons (i.e. spatial distribution or abundance of predators, or activity of an individual predator). Finally, environments may be shared among all study subjects (such as a change in day length through a breeding season), or they may be unshared and vary independently among subjects (such as local patterns of food supply).

When such environmental effects on an individual's phenotype are omitted in the statistical model, the variance in phenotype that they generate will be partitioned into the estimated variance components. However, it is unclear whether these different patterns of environmental variation affect the fitted variance components in a consistent and predictable manner. For example, environments can be autocorrelated over time in a variety of ways. Recently, Mitchell et al. (2019) suggested that temporal autocorrelation in environmental conditions should affect repeatability and plasticity; they examined autoregressive autocorrelation in the residual (i.e.

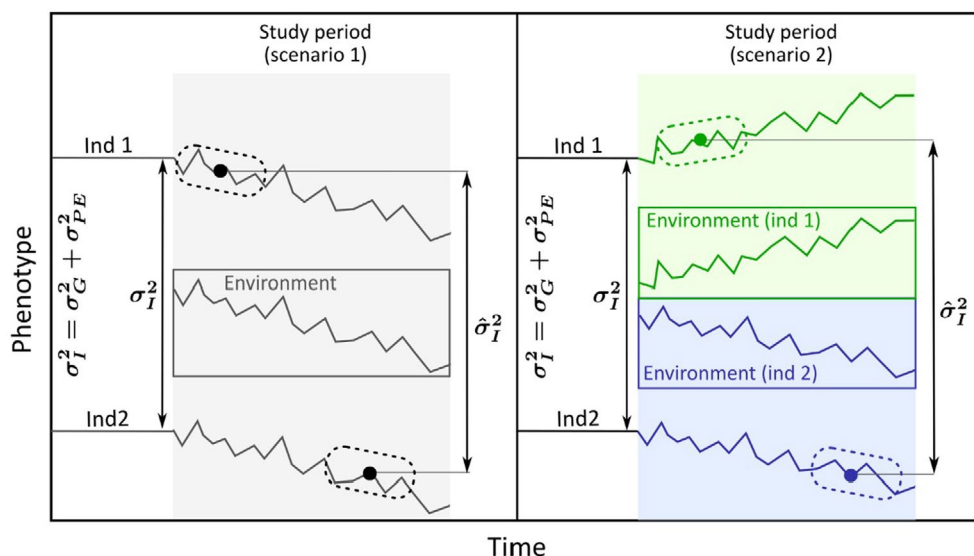


FIGURE 1 Illustration of the potential impact of biased sampling combined with an unmeasured environmental effect. At the start of the study period, individuals (ind 1 and ind 2) express some level of pre-study among-individual variance σ_I^2 in the trait, either through genetic variation (σ_G^2) and/or variation caused by permanent environmental effects, σ_{PE}^2 (i.e. environmental effects on the phenotypes that happened before the study period). In Scenario 1, individuals share the same changing environment during the study period (illustrated in the centre box), and their phenotypes are affected by this environment (assumed here to be with an $R^2=1$) within the period of study. However, as the two individuals are measured at different times, the effect of the unmeasured environment within the study generates additional among-individual variation. Similarly, in Scenario 2, individuals differ in phenotype entering the study, but here they experience independent environments (depicted in the two centre boxes, colour-coded for each individual). Again, if sampling occurs over a short period (dashed ovals) and varies among individuals (ind 1 compared to ind 2), whether the environment is shared (Scenario 1) or unshared (Scenario 2), the observed among-individual variance ($\hat{\sigma}_I^2$) from a model without any correction (e.g. null model, Table 1) is greater than the σ_I^2 due to the target processes occurring before the period of study.

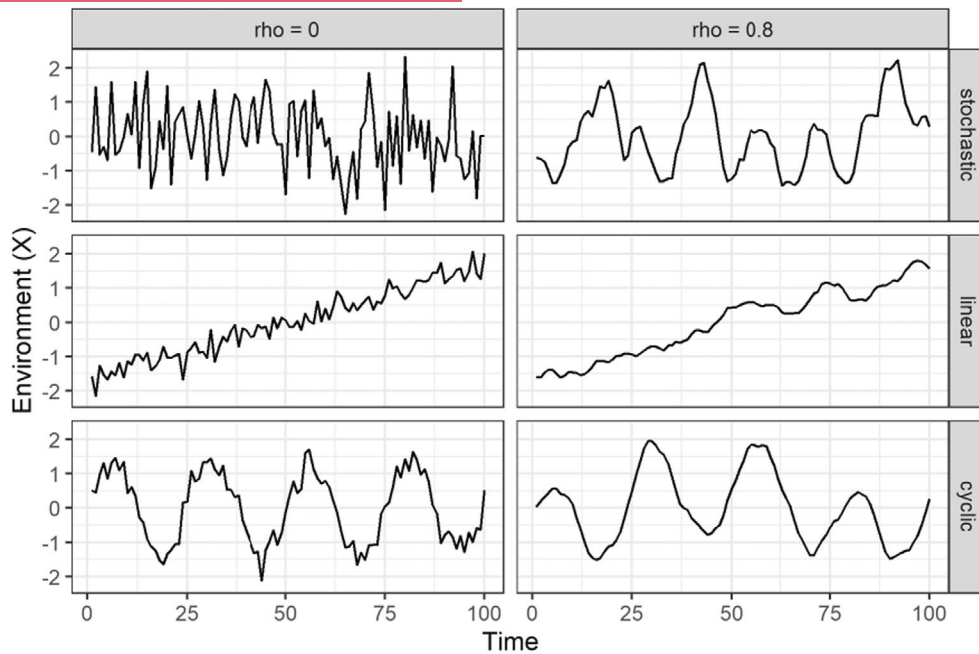


FIGURE 2 Examples of three simulated environmental effects (stochastic, stochastic with linear trends, and stochastic with cyclic patterns) without (correlation between two consecutive periods: $\rho=0$) and with additional autocorrelation ($\rho=0.8$) in minor fluctuations. See Table 1 for all details on the functions and parameter values used to simulate environmental effects.

within-individual) component of the trait, without varying sampling time. They found that, if uncorrected, this pattern could overestimate among-individual variance. They retrieved more relevant estimates of among-individual variance by adding an autoregression structure for time of sampling into the residuals, or by skipping some measurement times. This is as we should expect if the data analysis model (mixed model with autoregression on residuals) matched the data generation process (simulations) (Westneat et al., 2020). In nature, however, many additional and unknown variables acting at unknown time scales are not modelled or even modellable. For example, the environment could exhibit autocorrelation, while the organism ‘merely’ responds to the environment. While this process produces autocorrelation in residuals, it is not the only biological process that can do so. This means that directly generating autocorrelation in phenotype residuals, as Mitchell et al. (2019) did, could produce a mismatch between the biological process and analysis model, potentially leading to estimation issues (Westneat et al., 2020).

In this paper, we investigate patterns of environmental effects that, when combined with variation in the timing of sampling among groups or subjects (‘repeatability in timing of sampling’), might result in conflation of pre- and during-study effects. This can impact the interpretation of an analytical model. It is important to emphasize that we treat σ_i^2 as the among-individual variance before our simulated study (i.e. pre-study variance), and we are interested in the ability of data collected in realistic ways to retrieve that value. For this reason we consider the ‘extent of conflation’, $EC(\hat{\sigma}_v^2) = E(\hat{\sigma}_v^2) - \sigma_v^2$, where subscript v refers to the level of variance estimated (in our case among individuals or residual), and $E(\hat{\sigma}_v^2)$ is the expected value

of the estimated value $\hat{\sigma}_v^2$ of the pre-study variance σ_v^2 . The equation for EC is often used for measuring bias, but here we refer to it as extent of conflation, as our focus is on conflation of several types of biology.

We simulated datasets with known parameters and different sampling designs and showed how the extent of conflation on the estimates of variance components depends on the repeatability in the timing of sampling, the type of environmental effects (i.e. linear, cyclical, stochastic and shared or unshared among individuals) and the data analysis (vs. data generation) model. We then suggested a solution to correct for sampling bias in estimated variance components.

2 | METHODS

2.1 | Simulation models

We used the simulation squid R package (Allegue et al., 2017) to compare the performance of different mixed-effect models. Squid R used simulated datasets generated by the following phenotypic equation:

$$y_{ijk} = \beta_0 + I_k + \beta_1 x_{jk} + W_{ijk}, \quad (1)$$

where y_{ijk} is the phenotypic value of a trait of an individual k , period j and instance i , in the simulated population. Time j represents a period (e.g. a day, a week or a month) and regroups observations of the trait y for different individuals. β_0 is the population mean for the trait ($\beta_0=0$ in our simulations), I_k is the individual effect

TABLE 1 Analysis models used to fit simulated data in which an unmeasured environmental variable affects the trait and varies with some pattern over time.

Equation	No.	Model name	Description
$y_{ijk} = \hat{\beta}_0 + \hat{I}_k + \hat{W}_{ijk}$	2	<i>null</i>	Typical model run when a key predictor variable is omitted
$y_{ijk} = \hat{\beta}_0 + \hat{I}_k + \hat{\beta}_1 t_{jk} + \hat{W}_{ijk}$	3	<i>time.fix</i>	Time included as a fixed effect covariate
$y_{ijk} = \hat{\beta}_0 + \hat{I}_k + \hat{T}_j + \hat{W}_{ijk}$	4	<i>time.ran</i>	Time included as a random effect
$y_{ijk} = \hat{\beta}_0 + \hat{I}_k + \hat{T}_j + \hat{P}_{jk} + \hat{W}_{ijk}$	5	<i>time.ind.ran</i>	Time-by-individual included as a random effect
$y_{ijk} = \hat{\beta}_0 + \hat{I}_k + \hat{T}_{(ar1)j} + \hat{W}_{ijk}$	6	<i>ar1.time</i> or <i>ar1.time.ind^a</i>	Time as a fixed effect covariate but the model includes temporal autocorrelation

^aIn the case of an unshared environment, we specify an autoregressive process for each individual separately.

expressed as the deviation from the population mean and is characterized by a distribution $N(0, \sigma_y^2)$, β_1 is the population-mean effect of the environment (slope) ($\beta_1 = 1$ in our simulations). Finally, x_{jk} is the value of the environment x at period j for individual k , and W_{ijk} is the residual (i.e. within-individual) deviation associated with trait y measured at instance i , period j , and on individual k , and is characterized by a distribution $N(0, \sigma_W^2)$. Environmental variables x_{jk} are simulated as following a standardized normal distribution $N(0, 1)$. Note that for our analyses we assumed that x did not change within period j for individual k (hence x_{jk}). Also note that, for the need of some analyses, we simulated values of y at two instances i per individual k for each period j . These two values y_{ijk} thus differ in their values for W_{ijk} .

For each simulated population, we generated 100 individuals, each with two measures (instances) of the trait over five periods, resulting in 10 measures per individual over 100 total periods ($j \in [1, 100]$). We generated 100 populations per simulation scenario. Details about the values of the parameters used for the simulations are shown in Table S1.

The function *squidR* allows for the generation of different (types of) environmental variables affecting a focal phenotypic trait (Allegue et al., 2017). Here we chose to generate two scenarios with temporal autocorrelation in the environment and some temporal trends (either linear or cyclic) combined with the temporal autocorrelated process (Table S1). Furthermore, we were interested in the effects that shared versus unshared environments may have on the estimation of the among-individual variance ($\hat{\sigma}_I^2$) of the trait (Table 1). Note that only the temporal autocorrelation component in the environmental effect was unshared and generated separately for each individual; in contrast, the linear and cyclic trends were always shared among individuals.

Function *squidR* can generate phenotypic data for individuals following an equation (e.g. Equation 1), and sample among these data in a way analogous to field sampling by a biologist (Allegue et al., 2017). We can thus modify the sampling design and analyse

its impact on parameter estimation relative to true values. For this study, we varied the repeatability in the timing of sampling ($R_{IS}^2 \in [0, 1]$), which quantifies the consistency of individual sampling interval within the total sampling duration (i.e. 100 periods). Each individual's sampling interval was set as a fraction of the total sampling duration, equal to $1 - R_{IS}^2$, and randomly positioned within the overall timeframe. Samples for each individual were then randomly drawn from within their specific sampling interval. At one extreme, when $R_{IS}^2 = 0$, all individuals were sampled randomly across the total sampling duration (Figure 3). As R_{IS}^2 approached 1, individual sampling intervals became increasingly distinct, resulting in samples for each individual clustering tightly within a shorter interval (Figure 3; Table S1). To ensure that each individual was measured in at least five periods, we restricted R_{IS}^2 to values < 0.95 .

2.2 | Analysis models

When collecting data on real organisms, in practice, no research team can measure all the environmental variables affecting the phenotype of a trait. We thus evaluate the extent to which the repeatability in the timing of sampling (R_{IS}^2) impacts estimates of pre-study among-individual variance ($\hat{\sigma}_I^2$) and within-individual variance ($\hat{\sigma}_W^2$) when we omit the effect of the environment x in the mixed model (*null* model, Equation 2; Table 1):

$$y_{ijk} = \hat{\beta}_0 + \hat{I}_k + \hat{W}_{ijk}, \quad (2)$$

where $\hat{\beta}_0$ is the estimate of the population mean, $\hat{I}_k$ is the estimated deviation for individual k to the population mean, and $\hat{W}_{ijk}$ is the estimated residual (within-individual) value associated with y_{ijk} measured at instance i and time j on individual k .

In a second model, we followed the recommendations of Biro and Stamps (Biro & Stamps, 2015), who suggested that, lacking

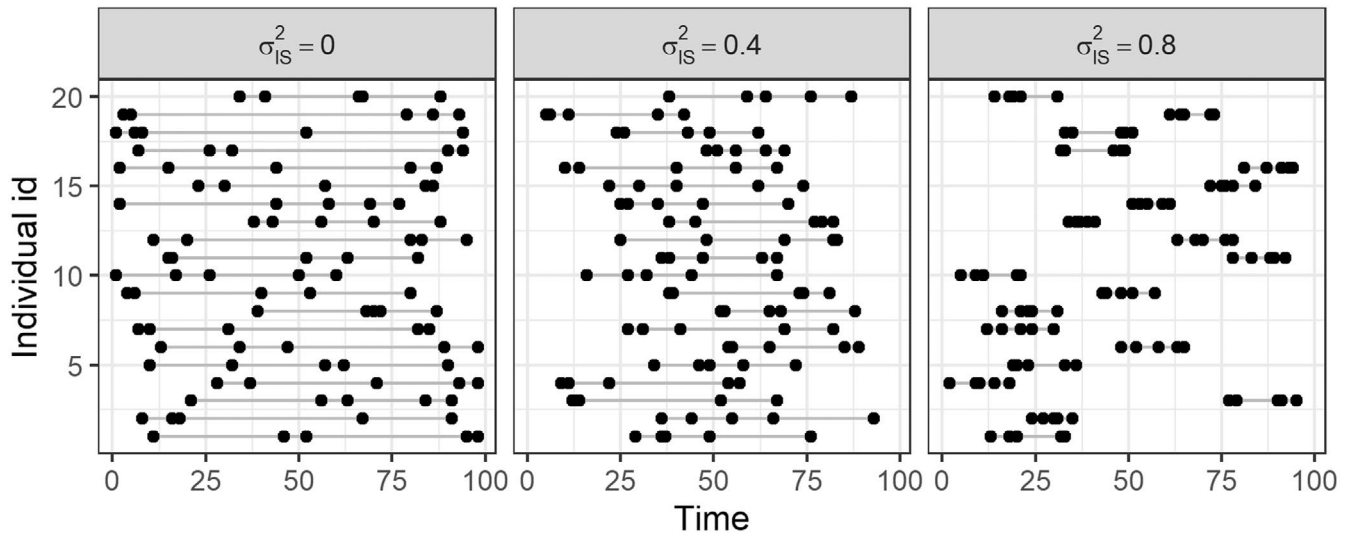


FIGURE 3 Examples of three sampling schemes for 20 individuals sampled five times each, for which the repeatability in the timing of sampling (R^2_{IS}) is 0, 0.4 and 0.8. High repeatability results in sampling periods frequently being disjunct among individuals.

information on x , one should fit the fixed effect time into the analysis model (*time.fix*, Equation 3):

$$y_{ijk} = \hat{\beta}_0 + \hat{l}_k + \hat{\beta}_1 t_{jk} + \hat{W}_{ijk}, \quad (3)$$

where $\hat{\beta}_1$ is the estimated coefficient of the slope of y on time and t_{jk} is the value of period j at which individual k is sampled. Equation (3) assumes that the mean of y_{ijk} follows a linear trend with time. Note, however, that for this model to fulfil its objective of using time as a proxy for x , it must implicitly assume that x and time are correlated, and therefore that y_{ijk} follows a linear trend with time and, therefore, with x .

In a third model, we included time as a random effect (in the case of our simulation, we have 100 instances, and thus each period forms a category) as an alternative model for solving the same issue (*time.ran*, Equation 4):

$$y_{ijk} = \hat{\beta}_0 + \hat{l}_k + \hat{T}_j + \hat{W}_{ijk}, \quad (4)$$

where $\hat{T}_j$ is the estimated deviation for time j at which individual k is measured versus the population mean. The rationale of this analysis model is that it could capture any form of temporal change in the environment, that is, both linear and nonlinear.

In a fourth model, we kept individual ($\hat{l}_k$) and time ($\hat{T}_j$) as random effects and we added as a new random effect the combination between individual and period (i.e. 'series' in Araya-Ajoy et al., 2015) (*time.ind.ran*, Equation 5):

$$y_{ijk} = \hat{\beta}_0 + \hat{l}_k + \hat{T}_j + \hat{P}_{jk} + \hat{W}_{ijk}, \quad (5)$$

where $\hat{P}_{jk}$ represent the individual-period effects. Concretely, as there were two observations per individual in each period j (see simulation models section), we could use three columns in our dataset to represent the three random effects: individual ID (e.g. individual A, B, C,

...), period (e.g. 1, 2, ...) and individual-period (A1, A2, B1, B2, C1, C2, ...). This model thus estimates four variance components: one variance among individuals, one variance among periods, one variance among periods within individuals and one variance within individual within period (residual). The rationale of this analysis model is that it may capture any form of temporal change specific to individuals (i.e. unshared (non) linear environmental effects).

Finally, in a fifth model, we followed the suggestion of Mitchell et al. (2019) and included a stationary autoregressive process of order 1 (AR1) (*ar1.time*, Equation 6):

$$y_{ijk} = \hat{\beta}_0 + \hat{l}_k + \hat{T}_{(ar1)j} + \hat{W}_{ijk}, \quad (6)$$

where $\hat{T}_{(ar1)j}$ is the estimate of the deviation at period j to the population mean, which is modelled as following a normal distribution $N(0, 1) \cdot e^{-\alpha \Delta_j}$, with $\alpha = \ln(\rho)$, ρ being the correlation between two periods and Δ_j being the interval between two periods (Table S1). In the cases where the environment was unshared among individuals, we structured the autoregressive process in $T'_{(ar1)j}$ by individual (*ar1.time.ind*). The models are summarized in Table 1.

All the models were run using the function *lmer* in package *lme4* (Bates et al., 2015), except Equation 6 (the autocorrelation model), which used the function *glmmTMB* in the package *glmmTMB* (Brooks et al., 2017).

3 | RESULTS

3.1 | Extent of conflation on the estimators in the null model

In the null model, whatever the repeatability in the timing of sampling (R^2_{IS}), $\hat{\sigma}^2_l$ was overestimated compared to σ^2_l and $\hat{\sigma}^2_w$ was underestimated compared to σ^2_w in all simulation scenarios (row A, in

Figures 4 and 5; Figures S1–S4). The extent of conflation on $\hat{\sigma}_I^2$ increased with the increase in R_{IS}^2 . However, this was associated with a reduction in the extent of conflation on $\hat{\sigma}_W^2$. In our simulations, $\hat{\sigma}_W^2$ included both simulated residual variation (fixed to 0.05) and the other residual variance that remained unexplained by the specific model. Thus, in the absence of conflating effects on $\hat{\sigma}_I^2$, all the variance caused by the unmeasured environmental effect is added to

the residual variance, and $\hat{\sigma}_W^2$ should be equal to 1.05. Therefore, the conflating effects on $\hat{\sigma}_I^2$ generally come from a transfer of variance from $\hat{\sigma}_W^2$ to $\hat{\sigma}_I^2$.

When the environment was stochastic or had a cyclic trend, as temporal autocorrelation increased, values of $\hat{\sigma}_W^2$ varied independently of $\hat{\sigma}_I^2$ (row A in Figure 4; Figure S1). We note that this excess variation in $\hat{\sigma}_W^2$ disappeared when the environment was

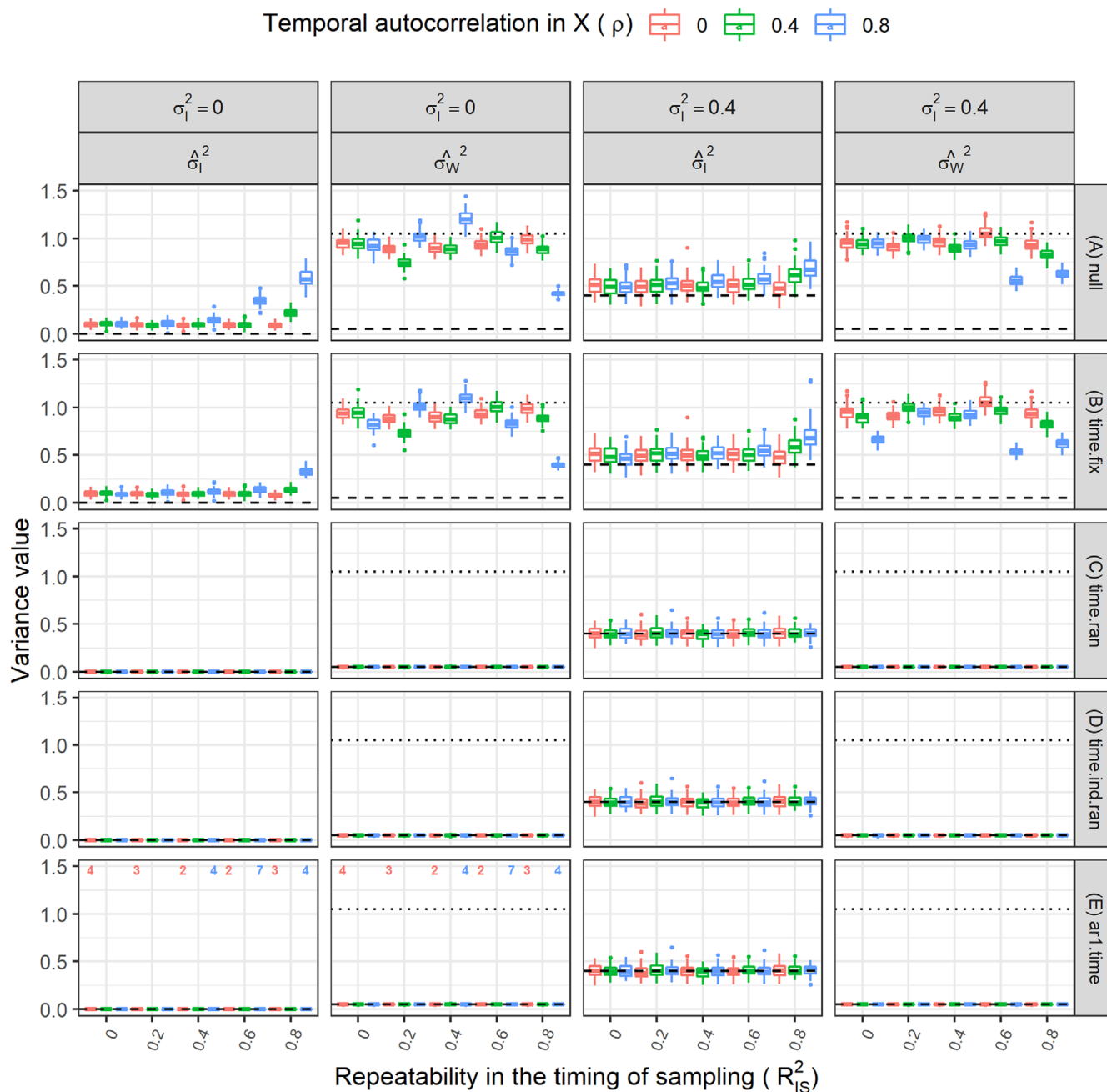


FIGURE 4 Results for the effect of a shared environment with temporal autocorrelation but no trend. Estimates of the among- ($\hat{\sigma}_I^2$) and within-individual ($\hat{\sigma}_W^2$) variances as a function of the repeatability in the timing of sampling (R_{IS}^2) for different values of σ_I^2 (0 and 0.4) and fitted models (rows). See the *Methods* section for a detailed description of the fitted models. The environmental effect (x) is simulated as stochastic and shared among individuals, with different values of autocorrelation (ρ). The horizontal dotted line, in the $\hat{\sigma}_I^2$ panels, indicates the true value of σ_I^2 . The bottom horizontal dotted line, in the $\hat{\sigma}_W^2$ panels, indicates the true simulated value of the residual variance and the upper dotted line indicates the true simulated value of the sum of the environmental variance and the simulated residual variance. The number of fitted models that did not converge is displayed on top of each simulation scenario if different from 0.

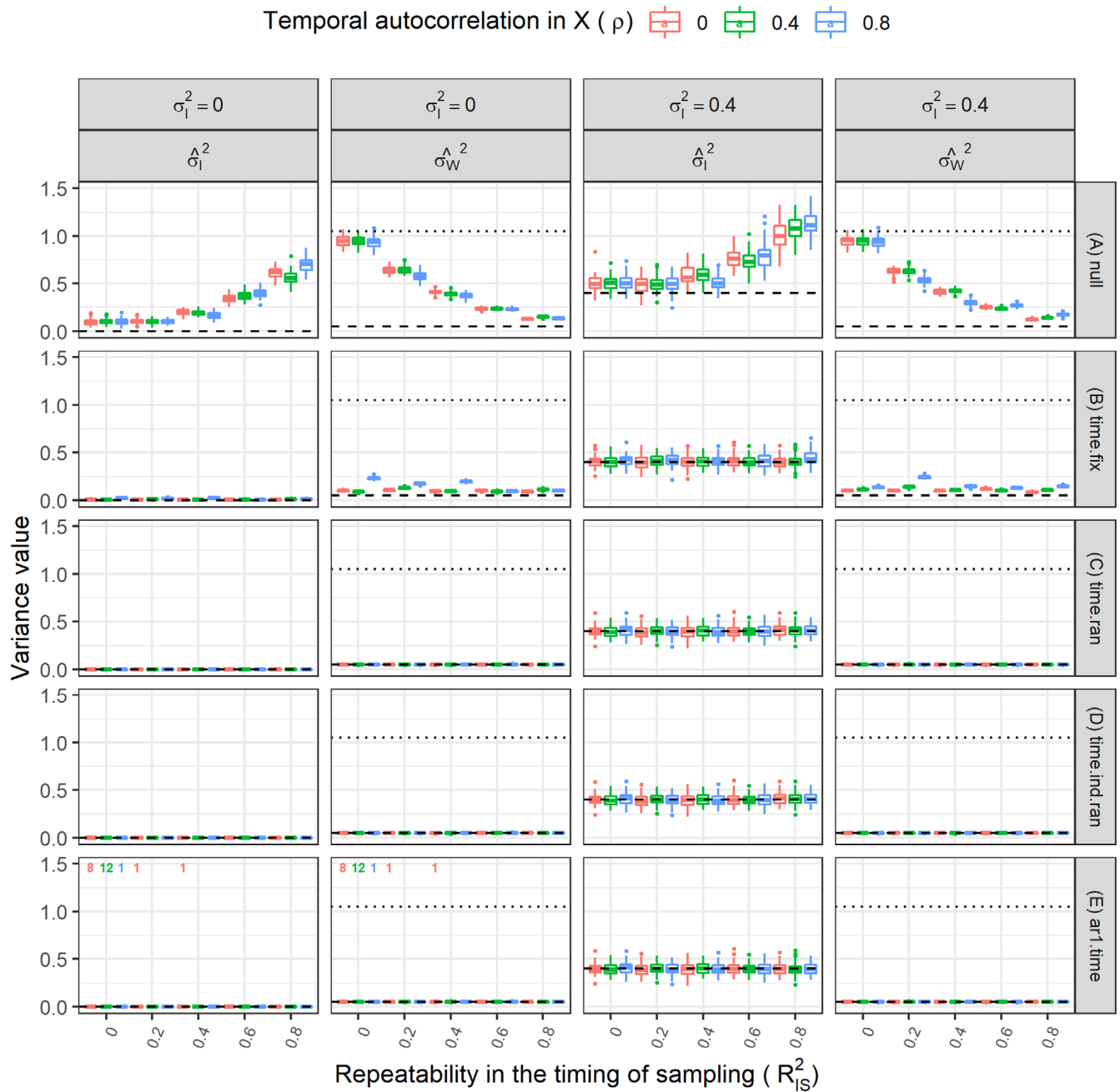


FIGURE 5 Results for the effect of a shared environment with temporal autocorrelation and a general linear trend. Estimates of the among- ($\hat{\sigma}_I^2$) and within-individual ($\hat{\sigma}_W^2$) variances as a function of the repeatability in the timing of sampling (R_{IS}^2) for different values of σ_I^2 (0 and 0.4) and fitted models (rows). See Section 2 for a detailed description of the fitted models. The environmental effect (x) is simulated as stochastic and shared among individuals, with different values of autocorrelation (ρ), and with a general linear trend. The horizontal dotted line, in $\hat{\sigma}_I^2$ panels, shows the true value of σ_I^2 . The bottom horizontal dotted line, in the $\hat{\sigma}_W^2$ panels, indicates the true simulated value of the residual variance, and the upper dotted line indicates the true simulated value of the sum of the environmental variance and the simulated residual variance. The number of fitted models that did not converge is displayed on top of each simulation scenario if different from 0.

unshared among individuals (row A in Figures S2 and S4). The extent of conflation on $\hat{\sigma}_I^2$ increased with R_{IS}^2 and the temporal autocorrelation when the environment was stochastic (row A in Figure 4; Figure S2). The extent of conflation on $\hat{\sigma}_I^2$ also increased only with R_{IS}^2 when the environment had a linear trend (row A in Figure 5; Figure S3). The extent of conflation on $\hat{\sigma}_I^2$ was relatively stable when the environment had a cyclic trend (row A in Figures S1 and S4).

3.2 | Comparison of model performance

When the environment was shared among individuals, the models *time.ran* (row C in Figures 4 and 5; Figure S1), *time.ind.ran* (row D) and *ar1.time* (row E) corrected for the extent of conflation on $\hat{\sigma}_I^2$ and $\hat{\sigma}_W^2$ in all simulation scenarios. Only in the presence of a linear trend in x did the model *time.fix* (row B in Figure 5) correct for the extent

of conflation on $\hat{\sigma}_I^2$ and reduce conflating effects in $\hat{\sigma}_W^2$, but it did not correct it when the environment was only stochastic (row B in Figure 4) or showed a cyclic trend (row B in Figure S1).

When the environment was unshared among individuals, no model could correct all the conflating effects on $\hat{\sigma}_I^2$ and $\hat{\sigma}_W^2$ in any of the simulation scenarios (Figures S2–S4). When the environment was stochastic and unshared (Figure S2), the models *time.ind.ran* (row D) and *ar1.time.ind* (row E) reduced the conflating effects on $\hat{\sigma}_I^2$ and corrected for these effects on $\hat{\sigma}_W^2$. The model *ar1.time.ind* corrected for the conflating effects on $\hat{\sigma}_I^2$ and $\hat{\sigma}_W^2$ when σ_I^2 is 0, but $\hat{\sigma}_I^2$ was underestimated when σ_I^2 was 0.4 and the extent of conflation increased with R_{IS}^2 and the temporal autocorrelation (row E in Figure S2).

When the environment was unshared and had a linear trend (Figure S3), all models in all simulation scenarios reduced the conflating effects on $\hat{\sigma}_I^2$ and $\hat{\sigma}_W^2$ but did not fully correct them. In this context, the models *time.fix* (row B) and *time.ind.ran* (row D) seem to perform better than the other models. The model *time.fix* corrected the conflating effects on $\hat{\sigma}_I^2$ but left some effects on $\hat{\sigma}_W^2$, whereas *time.ind.ran* corrected them on $\hat{\sigma}_W^2$ but induced an overestimation in $\hat{\sigma}_I^2$ at high values of R_{IS}^2 (0.8). The model *ar1.time.ind* (row E) estimated $\hat{\sigma}_I^2$ to 0 when σ_I^2 was 0.4 except for simulations, where R_{IS}^2 was 0.8 and the temporal correlation was 0 or 0.4, for which conflating effects increased $\hat{\sigma}_I^2$ (Figure S3).

When the environment was unshared and had a cyclic pattern (Figure S4), the models *time.ran* (row B) and *time.ind.ran* (row D) performed the best in reducing the conflating effects on $\hat{\sigma}_I^2$ and $\hat{\sigma}_W^2$, although *time.ind.ran* was better in reducing these effects in $\hat{\sigma}_W^2$. Similarly to other simulation scenarios, the model *ar1.time.ind* (row E) corrected the conflating effects on $\hat{\sigma}_I^2$ and $\hat{\sigma}_W^2$ when σ_I^2 was 0, but it also induced an underestimation of $\hat{\sigma}_I^2$ when σ_I^2 was 0.4, and these effects increased with R_{IS}^2 .

4 | DISCUSSION

Our simulations show that, in the presence of unmeasured environmental effects on a trait, repeatability in the timing of sampling (R_{IS}^2) can lead to an estimate of among-individual variance in the trait ($\hat{\sigma}_I^2$) that deviates from that expected if only pre-study processes were important. This effect of unmeasured environments also alters the estimate of within-individual variance ($\hat{\sigma}_W^2$) compared to values expected from random unmeasured 'noise'. The model including only individual as a random effect and an intercept as a fixed effect (i.e. the *null* model) shows that these deviations increase with R_{IS}^2 and are exacerbated by the strength of the temporal autocorrelation in the unmeasured environmental effect on the trait. We can therefore confirm our intuition that if individuality exists in both pre-study and within-study events, the model's parameters will reflect the total. This compromises the interpretation of random intercept variation.

These problems appear regardless of whether environments are shared or unshared. In the case of shared environments, they can be

dealt with easily by specifying a random effect structure with period (i.e. *time.ran*, *time.ind.ran* or *ar1.time.ind*) in the model. But this problem becomes intractable when the environment is unshared, as no solution we considered completely corrected it. As we have little understanding of how often important environmental effects on the trait are shared or unshared between individuals, our results suggest that the magnitude of pre-study among-individual variance, repeatability or heritability, is likely to be overestimated in many studies, leading to inaccurate conclusions about consistent differences between individuals. Of course, an intuitive solution would be to include as fixed effects in the model the environmental variables we suspect could affect the studied trait. In wild populations, however, we always run the risk of missing an important environmental variable affecting that trait.

4.1 | Efficiency of different models in correcting the issue caused by repeatability in the timing of sampling

We tested potential statistical solutions to correct the conflation of pre-study and within-study effects affecting $\hat{\sigma}_I^2$ and $\hat{\sigma}_W^2$, with modest success. Performance depends on the model (see Table 1) and the type of environmental effect. Biro and Stamps (2015) suggested including time as a fixed effect (in our models: period as a covariate). Our results indicate that this solution only works when the unmeasured environmental effect shows a temporal linear, or cyclic trend. For example, this model could control a situation where temperature or precipitation changes throughout the season (Clermont et al., 2018), or across years (Charmanier & Gienapp, 2013; Nussey et al., 2005). In the presence of stochastic environmental changes, especially with unshared environmental effects, the model can also produce estimates that are highly affected by the conflation of pre-study and within-study effects.

Mitchell et al. (2019) proposed and found that including an autocorrelation (AR1) function alters variance components when temporal autocorrelation seems likely. Our results differ from theirs. In their simulations, Mitchell et al. (2019) assumed that any environmental effects on the trait that were autocorrelated were found in the residual portion of the total phenotypic variance. Hence, a mixed model that includes an autocorrelation function (i.e. *ar1.time.ind*) can exactly estimate the part of the phenotypic variance caused by this effect and correct for the bias in $\hat{\sigma}_I^2$ (i.e. generation matched analysis). By contrast, using *squidR* (Allegue et al., 2017), we simulated temporal autocorrelation in the environmental effect x , then let the phenotype of trait y change as a function of x . Initially, these two approaches may appear to generate similar biological phenomena, but studies have shown that mixed models can only appropriately fit data structures generated based on the same assumptions regarding variance components (Westneat et al., 2020). Our approach simulates temporal autocorrelation in the environment directly, with the plastic response reflecting that autocorrelation, a situation we think is common

in natural systems. Using a simple autocorrelation function (e.g. AR1) is not modelling the biological process that generated the response variable and will thus not always provide correct estimates of variance components. Autocorrelation in the environment can produce among-individual differences, but if temporal autocorrelation in y depends on intrinsic changes in the animal state, as proposed by Mitchell et al. (2019), such as if behaviour depends on body condition, then all the autocorrelated changes will occur within individuals, in which case using the AR1 function may produce appropriate estimates. Mitchell et al. (2019) also recommended sampling individuals at large enough lag steps. What might be a sufficient lag is unclear, however, if one is concerned about unknown environmental effects. We note here that this situation illustrates the idea that a deep understanding of statistical models is necessary to improve our understanding of biology and vice versa.

When individual phenotypes were affected by environmental effects that were shared by all the individuals, regardless of time course, including period as a random effect (e.g. Wilson et al., 2010) corrected for the conflation of effects on both $\hat{\sigma}_I^2$ and $\hat{\sigma}_W^2$, as the portion of variance caused by the unmeasured environmental effects goes into the estimate of variance for period. We thus recommend that researchers use period (e.g. day or week) as a random effect rather than as a fixed effect to control the conflation of effects on the variance estimates. However, when each individual experienced idiosyncratic (unshared) environmental effects, and the repeatability in the timing of sampling (R_{IS}^2) was higher than 0.6, none of the models could correct for the conflation of effects on $\hat{\sigma}_I^2$. Here the solution does not rely on defining a specific statistical model, but on a sampling or resampling plan that reduces R_{IS}^2 as much as possible. Focusing on a subsample with reduced R_{IS}^2 might at least allow one to identify how big a problem might exist in the full dataset.

A model including individual-period combinations as a random effect (model *time.ind.ran* [Equation 5] in Table 1) was the most appropriate solution to deal with potential temporal sampling issues. This model can be fit easily if the data exhibit a temporal hierarchical structure characterized by periods of intensive data collection separated by periods of no data collection (Araya-Ajoy et al., 2015; Mitchell et al., 2019). Take, for example, a dataset with individuals measured on several occasions (i.e. equivalent to instances in our simulation model) each spring (i.e. equivalent to periods) over several years, but different individuals are sampled across different years. In such a situation, we can include an individual-year combination as an additional random effect (see model *time.ind.ran* [Equation 5] in Table 1) to estimate variance among individuals, variance within individuals among years, and variance within individuals within years (Araya-Ajoy et al., 2015; Clermont et al., 2019). We thus advise that the trait be measured repeatedly within each individual, within each period and across different periods. A suboptimal, but sometimes helpful strategy is to restructure that dataset at the analysis stage to obtain periods

of intensive data collection separated by periods of no data collection. That may mean losing some data to create different temporal blocks of repeated measures to reduce R_{IS}^2 . Note that, as with the same example as above, if individuals were measured once every spring over several years, one may create pairs of years as periods to obtain two measures per period. This solution is not perfect, as some unmeasured environmental changes may still exist within each block, and thus this solution reflects a best-of-a-bad-job compromise.

4.2 | Conflation of effects caused by spatially heterogeneous environment

The blending of processes affecting random intercept variance may also be due to spatial heterogeneity. Even if individuals are sampled within the same period, we expect unshared environmental effects to add to the among-individual variance of the trait if they experience different environmental conditions because they use different habitats (Niemela & Dingemanse, 2017). If several individuals are measured at each location, including location (appropriately binned), as a random effect may correct for this problem. In the case of permanent environmental effects, individuals may already differ in their phenotype at the onset of the study, and a part of their σ_I^2 would be caused by these pre-study environmental effects (Figure 1). If these habitats continue to affect the phenotypes of individuals differently throughout the study (e.g. Scenario 2 in Figure 1), it may be necessary to include a random slope with period to control for the effects of repeatability in the timing of sampling. Note, however, that we have not tested the use of random slopes as a solution to sampling problems.

To our knowledge no study has applied the equivalent of the *time.ind.ran* approach to space. This approach would consist of including an individual-location combination as an additional random effect to estimate variance among individuals, variance within individuals among locations, and variance within individuals within locations. This analysis requires individuals to be sampled repeatedly within locations, and in different locations. This approach will be impossible if each individual remains in one location. Furthermore, because individual animals are not ubiquitous, this adds to the complexity of the model, as the individual-location combination will overlap with the individual-period combination. Other methods are available for controlling spatial autocorrelation effects on repeatability or heritability estimates (e.g. Stopher et al., 2012; van der Jeugd & McCleery, 2002), although they only capture the autocorrelation in the residual part, and for now it is unclear to what extent these methods are immune to the issues we raise in this paper. Caution is necessary when correcting for such effects, as individuals may use different habitats based on their phenotypic differences for the studied traits (i.e. matching habitat choice: Edelaar et al., 2008; Edelaar & Bolnick, 2019, or genotype-environment correlation: Gervais et al., 2022; Saltz &

Nuzhdin, 2014). In such cases, correcting for location may unduly remove a significant part of the among-individual variance in the trait (Wilson, 2018), and experimental approaches (i.e. translocation experiments) might be required (Niemela & Dingemanse, 2017).

4.3 | Some guidelines

Biologists measuring real organisms need guidelines for how to deal with the inevitable fact that not all environmental effects can be accounted for. We recommend that researchers first control the temporal sampling design of their study to reduce R_{IS}^2 as much as possible. For example, one can try to sample as many individuals as possible in successive temporal blocks separated by longer periods. In field conditions, where not all the individuals are available and captured or observed on demand. We recommend that researchers estimate and report R_{IS}^2 in publications. This can be done by running a mixed model on the date of sampling, using individual or group identities as a random effect. Our results show that when $R_{IS}^2 < 0.6$ researchers may feel confident that unmeasured environmental effects will not have strong effects on the estimation of the components of variance. However, the value of 0.6 is not generalizable to all situations and R_{IS}^2 also depends on the unknown population-mean effect of the environment (β_1). Thus, as a rule of thumb, any non-negligible value of R_{IS}^2 should raise the attention of researchers and make them cautious in interpreting their results. Researchers who could not reduce R_{IS}^2 through planning to sample their random effect units, could resample data to reduce R_{IS}^2 . For example, they may remove data sampled very early or tardy during the study to reduce R_{IS}^2 . They could also use a temporal block design to reduce R_{IS}^2 and avoid biasing estimates of variance components. In the case of large, long-term datasets, which cover several generations of individuals, running models for different periods would probably reduce R_{IS}^2 .

None of the models we proposed provided satisfactory solutions when individuals experienced unshared environmental effects and high R_{IS}^2 . In some studies, it is almost impossible to minimize R_{IS}^2 . For example, in species with non-overlapping generations, or populations studied over several decades, observations from different individuals will cluster in time. This is particularly important for studies analysing among-individual variation in plasticity in response to climate change or other anthropogenic changes from long-term datasets (Charmanier & Gienapp, 2013; Husby et al., 2010; Nussey et al., 2005; Phillimore et al., 2010). Although we have not run simulations illustrating this scenario, we can conceive that conflation of pre-study and within-study effects could occur at both among-individual and additive genetic variance levels. For example, when used on long-term datasets of wild populations, the animal model (Wilson et al., 2010) might be subject to similar conflating effects on estimates of additive genetic variance $\hat{\sigma}_A^2$ if the R_{IS}^2 among different lineages is high. Estimating among-group variance (i.e. families, groups, sites, populations) will

suffer from the same issues if these groups are sampled at different periods. For quantitative genetic studies on long-term datasets of wild populations, period (i.e. year) is traditionally included as a random effect to correct for such conflating effects (Wilson et al., 2010). However, as our results show, in situations where individuals/lineages experienced unshared environmental effects, time as a random effect is not sufficient to avoid overestimation. Also note that some types of groups, such as populations or sites, will have a higher probability of being affected by unshared environmental effects, which may lead to stronger issues. In such cases, if data permit it, it might be worth estimating repeatability or heritability within successive periods within a study to reduce R_{IS}^2 as much as possible.

5 | CONCLUSIONS

Our analyses show that, when units (e.g. individuals) are sampled non-randomly through time (i.e. increasing R_{IS}^2), misspecifying the model can lead to estimates of variance components that contain elements from multiple processes, confusing proper interpretation. We recommend researchers clarify the goals of their models and carefully design their sampling to reduce R_{IS}^2 as much as possible. However, in many circumstances data have already been collected or field conditions limit the possibility of sampling many individuals during short periods, raising caution with the interpretation of the variance components estimated. Including period or using a hierarchical time structure as a random effect will often correct for the conflation of effects. Researchers could also resample their dataset to reduce R_{IS}^2 , albeit at the cost of losing data.

The autocorrelation of the environment $\times$ exacerbated the impact of non-random sampling. We found no published information on the distribution of the autocorrelation coefficient for environmental variables such as temperature, precipitation, or resource abundance. Thus, at this stage, we can only recommend researchers to be conscious of the potential issue and to evaluate the autocorrelation coefficient of their traits if this pattern reflects the autocorrelation of the environmental variables affecting the phenotypes. We also encourage researchers to investigate environmental variables that may affect their study subject, for example, by analysing the temporal and spatial correlation structure of various variables to assess environmental redundancy. In the presence of strong environmental redundancy, using a few representative variables could limit sampling bias issues in the estimates of variance components.

We also have limited our study to normally distributed traits, and we warn readers of the risk of extrapolating our results to other types of distribution. We have not evaluated the impact of non-random sampling on the estimation of covariance between traits in our simulations; it is conceivable that they will suffer similar conflating effects. More importantly, the covariance could be overestimated or underestimated, depending on the direction of the reaction norms of the two traits analysed, and thus it would not be possible to predict the effect on the covariance estimates. To avoid

these problems, we can only recommend that readers limit R^2_{IS} as much as they can in their studies.

AUTHOR CONTRIBUTIONS

Denis Réale and Niels J. Dingemanse co-organized the project and wrote the first draft of the manuscript. Hassen Allegue contributed to the design of the study, coded and ran the simulations and commented on the successive drafts of the paper. All authors contributed by sharing ideas on the design, discussing the results and revising the manuscript.

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CONFLICT OF INTEREST STATEMENT

The authors declare having no conflict of interest.

PEER REVIEW

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DATA AVAILABILITY STATEMENT

All R codes used for the simulations presented in this article are available at <https://doi.org/10.5281/zenodo.17544120> (Allegue, 2025).

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

Table S1: Description and values of the simulation parameters.

Figure S1: Estimates of the among- ($\hat{\sigma}_I^2$) and within-individual ($\hat{\sigma}_W^2$) variances as a function of the repeatability in the timing of sampling (R_B^2) for different values of σ_I^2 (0 and 0.4) and fitted models (rows).

Figure S2: Estimates of the among- ($\hat{\sigma}_I^2$) and within-individual ($\hat{\sigma}_W^2$) variances as a function of the repeatability in the timing of sampling (R_B^2) for different values of σ_I^2 (0 and 0.4) and fitted models (rows).

Figure S3: Estimates of the among- ($\hat{\sigma}_I^2$) and within-individual ($\hat{\sigma}_W^2$) variances as a function of the repeatability in the timing of sampling (R_B^2) for different values of σ_I^2 (0 and 0.4) and fitted models (rows).

Figure S4: Estimates of the among- ($\hat{\sigma}_I^2$) and within-individual ($\hat{\sigma}_W^2$) variances as a function of the repeatability in the timing of sampling (R_B^2) for different values of σ_I^2 (0 and 0.4) and fitted models (rows).

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